

Tianyi ZHANG

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OBJECTIVE

Seeking semiconductor engineering opportunities at **AMD Shanghai**, with a focus on SoC verification, automation infrastructure, AI-assisted EDA tooling, and hardware/software workflow optimisation.

EDUCATION

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| MSc Precision Scientific Instrumentation
Nanyang Technological University | Expected 2026 – 2027
Singapore |
| <ul style="list-style-type: none">Focus on semiconductor instrumentation, precision electronics, sensing systems, and advanced hardware integration. | |
| BEng Mechanical Engineering (<i>First Class Honours</i>)
University College London | Sept 2023 – Jun 2026
London, United Kingdom |
| <ul style="list-style-type: none">First Class Honours, approximately equivalent to GPA 3.7+/4.0 in the Chinese grading system.Relevant coursework: control systems, simulation and modelling, numerical methods, data structures, embedded systems. | |

SKILLS

- Programming:** Python, Java, C++, MATLAB, Shell, Verilog
- Semiconductor & Verification Tools:** Synopsys VCS, Verdi, Linux, Perforce, RTL automation, regression testing
- Software & Data Systems:** FastAPI, Streamlit, SQL, Git, data processing pipelines, statistical forecasting
- Engineering Tools:** MATLAB Simulink, LabVIEW, ANSYS, Fusion 360

WORK EXPERIENCE

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| Assistant Project Engineer, MSIP Department
Advanced Micro Devices, Inc. (AMD) | Jul 2024 – Present
Shanghai / London |
| <ul style="list-style-type: none">Contributed to memory read/write IP development for AMD accelerator projects including MI350 and MI400.Developed automation and AI-assisted verification scripts for large-scale SoC signal inspection and traffic analysis.Built automated RTL error-injection frameworks to improve verification robustness and functional coverage evaluation.Maintained regression and validation workflows using Linux, Shell scripting, Python, Perforce, Synopsys VCS, and Verdi.Improved internal AI scripting tools by integrating intelligent command correction features to reduce user input errors. | |

PROJECTS

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|--|-----------------|
| Full-Stack Immigration Data Management System <i>FastAPI / Streamlit / SQL</i> | 2025 |
| <ul style="list-style-type: none">Developed a full-stack data management platform for the Singapore Immigration & Checkpoints Authority (ICA).Designed backend APIs and database structures using FastAPI for traveller record management and analytics.Built Streamlit dashboards for operational monitoring, data interaction, and visual analysis.Implemented short-term forecasting models to estimate future inbound traveller volume trends.Integrated automated data-processing pipelines to support decision-making workflows. | |
| Energy-Aware Route Optimisation for Electric Scooters <i>OSM / Graph Algorithms</i> | 2025 – Jun 2026 |
| <ul style="list-style-type: none">Developed a simulation framework for electric scooter energy modelling using real-world map and elevation data.Implemented graph-based routing algorithms with custom energy-consumption edge weights derived from terrain and traffic conditions.Designed a hybrid optimisation strategy combining regenerative braking analysis with energy-aware route planning.Built automated simulation pipelines for large-scale route evaluation and statistical comparison across urban scenarios. | |

Path Planning for Trailer Obstacle Avoidance | *A* Algorithms / Kinematic Constraints*2022 – 2023

- Developed a constrained path-planning framework for articulated trailer systems using an enhanced A* search algorithm.
- Integrated non-holonomic vehicle dynamics, steering continuity constraints, and collision-envelope detection into the planning heuristic.
- Generated dynamically feasible obstacle-avoidance trajectories while balancing computational efficiency and path optimality.

LEADERSHIP & ACTIVITIES

UCL Racing Formula E Team

Nov 2023 – Jun 2026

- Assisted in debugging motor signals and sensor feedback systems for the Formula E race car.

Vice President & Director of Program Control, Robotics Club

Sept 2020 – Jun 2023

- Led the construction and debugging of robotic control modules and competed in the FRC China Regional Competition.

HONORS & AWARDS

British Physics Olympiad Gold Award

Nov 2022

ADDITIONAL INFORMATION

Languages: Chinese Native, English Professional Working Proficiency

Interests: Football, Table Tennis, Reading, Travelling